

Hardware Pin Connection Document

Project: IoT-Based Multi-Sensor Monitoring System

LEDs, Buzzer, and Push Button

Component	Pin / Signal	Connected To	ESP32 Pin	Notes
Red LED	Anode (+)	ESP32	GPIO25	Via 220Ω resistor
Red LED	Cathode (−)	GND	GND	Common ground
Yellow LED	Anode (+)	ESP32	GPIO26	Via 220Ω resistor
Yellow LED	Cathode (−)	GND	GND	
Green LED	Anode (+)	ESP32	GPIO27	Via 220Ω resistor
Green LED	Cathode (−)	GND	GND	
Buzzer	Positive (+)	ESP32	GPIO23	Digital output
Buzzer	Negative (−)	GND	GND	
Push Button	Pin 1	ESP32	GPIO19	INPUT_PULLUP
Push Button	Pin 2	GND	GND	Active LOW

DHT11 (Temperature & Humidity Sensor)

Component	Pin / Signal	Connected To	ESP32 Pin	Notes
DHT11	VCC	3.3V	3V3	Power supply
DHT11	DATA	ESP32	GPIO13	Requires pull-up resistor
DHT11	GND	GND	GND	

10kΩ Resistor	One side	DATA line	GPIO13	Pull-up
10kΩ Resistor	Other side	3.3V	3V3	Signal stabilization

MQ Gas Sensors (MQ2, MQ4, MQ7)

Component	Pin / Signal	Connected To	ESP32 Pin	Notes
MQ2	VCC	5V	5V	High current device
MQ2	GND	GND	GND	Common ground
MQ2	AOUT	ESP32	GPIO34	Analog input
MQ4	VCC	5V	5V	Methane sensor
MQ4	GND	GND	GND	
MQ4	AOUT	ESP32	GPIO35	Analog input
MQ7	VCC	5V	5V	CO sensor
MQ7	GND	GND	GND	
MQ7	AOUT	ESP32	GPIO32	Analog input

Ultrasonic Sensor (HC-SR04)

Component	Pin / Signal	Connected To	ESP32 Pin	Notes
HC-SR04	VCC	5V	5V	Requires 5V
HC-SR04	GND	GND	GND	
HC-SR04	TRIG	ESP32	GPIO5	Output trigger
HC-SR04	ECHO	Voltage Divider	GPIO18	Must reduce voltage

Voltage Divider (ECHO Protection Circuit)

Component	Connection	Connected To	ESP32 Pin	Notes
Resistor (1kΩ)	Top resistor	ECHO → Junction	—	First stage
Resistor (1kΩ)	Middle resistor	Junction → Mid node	—	Series connection
Resistor (1kΩ)	Bottom resistor	Mid node → GND	—	Voltage drop
Junction Point	Output node	ESP32	GPIO18	Safe ~3.3V signal

MAX30100 / MAX30102 (Pulse + SpO₂ Sensor)

Component	Pin / Signal	Connected To	ESP32 Pin	Notes
MAX3010x	VIN	3.3V	3V3	Do NOT use 5V
MAX3010x	GND	GND	GND	
MAX3010x	SDA	ESP32	GPIO21	I2C bus
MAX3010x	SCL	ESP32	GPIO22	I2C bus

MPU6050 (Motion Sensor)

Component	Pin / Signal	Connected To	ESP32 Pin	Notes
MPU6050	VCC	3.3V	3V3	Do NOT use 5V
MPU6050	GND	GND	GND	
MPU6050	SDA	ESP32	GPIO21	Shared I2C
MPU6050	SCL	ESP32	GPIO22	Shared I2C

SSD1306 OLED Display

Component	Pin / Signal	Connected To	ESP32 Pin	Notes
SSD1306	VCC	3.3V	3V3	Do NOT use 5V
SSD1306	GND	GND	GND	
SSD1306	SDA	ESP32	GPIO21	Shared I2C
SSD1306	SCL	ESP32	GPIO22	Shared I2C

I2C Bus (Shared Connection)

Bus Line	Connected Devices	ESP32 Pin	Notes
SDA	MAX3010x, MPU6050, SSD1306	GPIO21	Shared data line
SCL	MAX3010x, MPU6050, SSD1306	GPIO22	Shared clock line

Important Notes

- All components share a **common GND**
- ESP32 is **3.3V logic device**
- **Voltage divider ($3 \times 1k\Omega$)** protects GPIO18
- MQ sensors require **5V + warm-up time**
- I2C devices are connected **in parallel**
- Ensure proper connection at the **junction point**